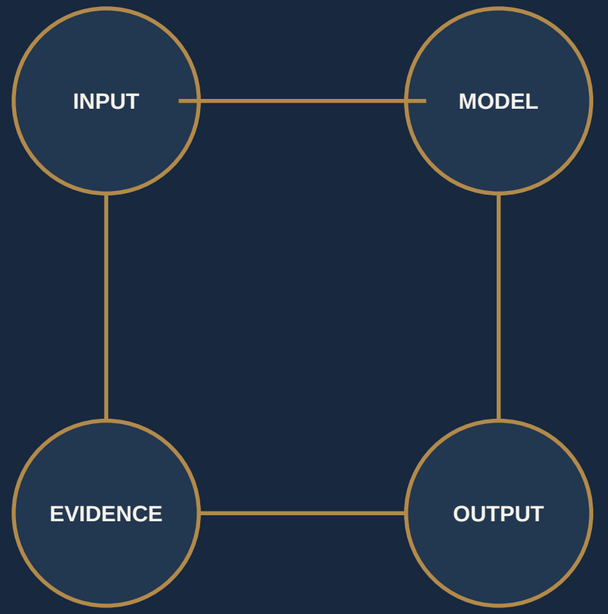
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Virtual Reality

AI.301 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define vr hardware.
- Apply key methods including tracking.
- Build or analyze a practical system involving 3d environments.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

VR hardware

02

METHODS

Tracking

03

SYSTEMS

3D environments

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · VR hardware

VR hardware

02 · Tracking

Tracking

03 · 3D environments

3D environments

04 · Interaction

Interaction

05 · Rendering

Rendering

06 · Locomotion

Locomotion

07 · Immersion and applications in

Immersion and applications in robotics and AI

08 · Mandatory practical: Build an

Mandatory practical: Build an interactive VR environment and demonstrate robot/AI interaction inside it

Core ideas, made explicit

AI.301 • Distinguish definitions, assumptions and evidence

- VR hardware
- Tracking
- 3D environments
- Interaction
- Rendering

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.301 • Distinguish definitions, assumptions and evidence

- Locomotion
- Immersion and applications in robotics and AI
- Mandatory practical: Build an interactive VR environment and demonstrate robot/AI interaction inside it

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving vr hardware.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using vr hardware.
- Implement or simulate a baseline using tracking.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.